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AI Workshop 01:

Linear Algebra for AI - Part 1

<h4> Understanding how machines see data</h4>

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Agenda

01. How machines represent data

02. Vectors: describing one object

03. Matrices: representing datasets

04. Basic operations on data

05. Reacap

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01. How machines represent data

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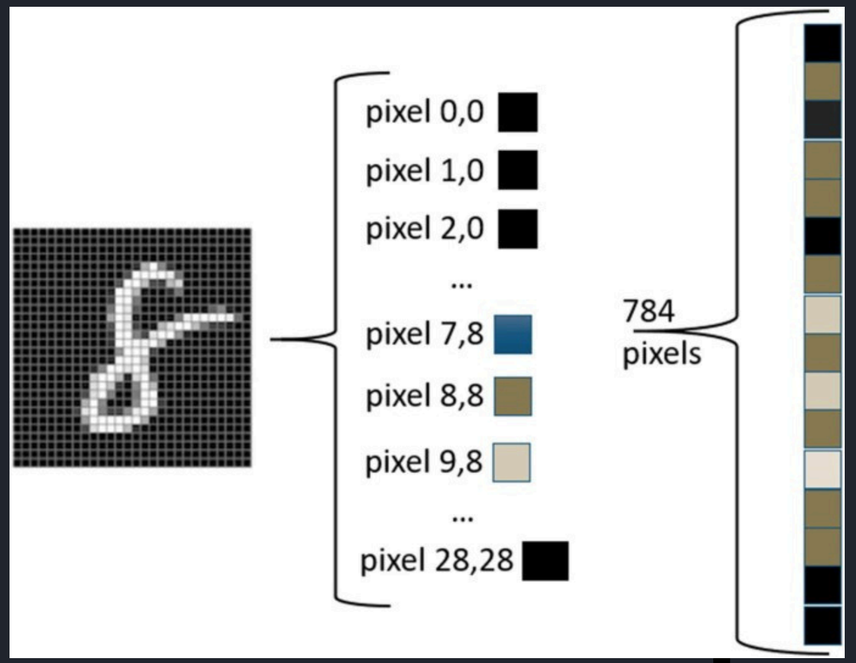
The Big Idea

No matter what the data is:

- Photo
- Sentence
- Person
- Stock price

the computer cannot see them like we do

👉 It can only see lists of numbers



The machine sees:

[0.1, 0.8, 0.3, 0.0, 0.9, ...]

A 28×28 image = 784 pixels

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A 4-dimensional embedding

cat =>	1.2	-0.1	4.3	3.2
mat =>	0.4	2.5	-0.9	0.5
on =>	2.1	0.3	0.1	0.4
...	...			

The word "cat" might become:

[0.2, -1.3, 0.7, 4.1, ...]

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Items

Users

10	-1	8	10	9	4
8	9	10	-1	-1	8
10	5	4	9	-1	-1
9	10	-1	-1	-1	3
6	-1	-1	-1	8	10

User-item Interaction matrix

👉 Again, one point in space

What You Must Leave With

📌 By the end of this session you can:

- Read shapes like: X ∈ R^{100×10}
- See a dataset as a matrix
- Interpret what a transformation does
- Think geometrically about data

Not: Solve equations Yes: Interpret models



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02. VECTORS

What Is a Vector?

📌 A vector is:

A list of numbers that describes one object

Pandas DataFrame from_dict()

```
{
 'name' : ['Joe', 'Nat', 'Harry'],
 'age'  : [20, 21, 19],
 'marks': [85.1, 77.8, 91.54]
}
```

	name	age	marks
0	Joe	20	85.1
1	Nat	21	77.8
2	Harry	19	91.54

A student:

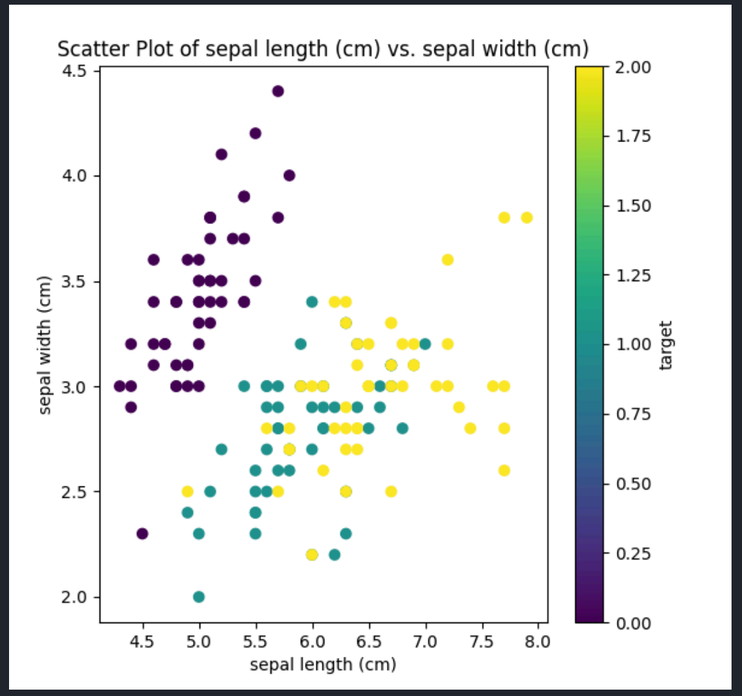
- [Name, Age, Marks]
- [Joe, 20, 85.1]

One object

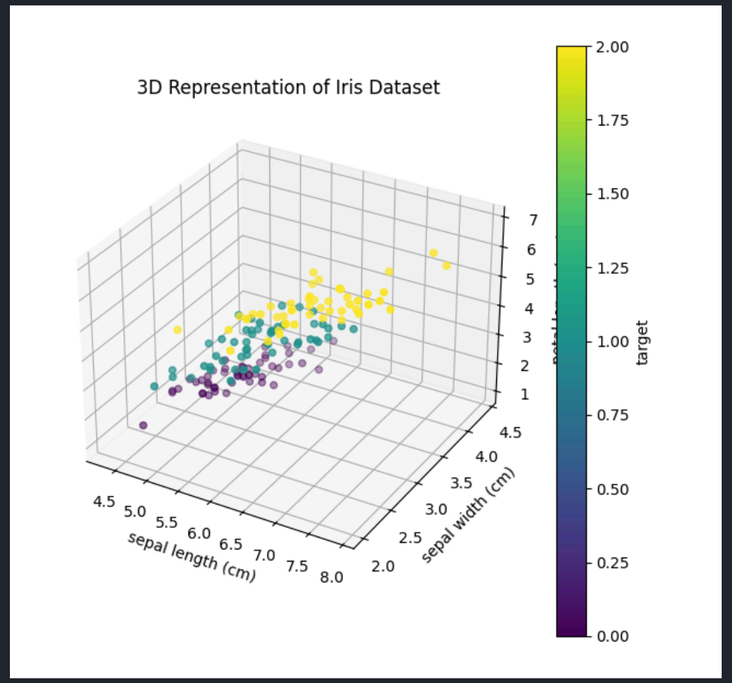
👉 Many features...

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Vector as Point in Space



📌 2 features → point in 2D



📌 3 features → point in 3D

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Why space?

Because each number is a dimension



If you have:

- 2 numbers → 2D space (like a plane)
- 3 numbers → 3D space
- 100 numbers → 100D space (you can’t draw it, but it exists mathematically)

every data point lives somewhere in a big invisible space

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Dimension

Dimension = number of features



Examples (previous images) :

- Image (28×28) → 784 dimensions
- Word embedding → 300 dimensions
- User profile → 20 dimensions

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Norm (Magnitude)

Norm = how “big” the vector is



Interpretations:

- Strength of signal
- Importance
- Intensity

Examples:

- Loud vs quiet sound
- Bright vs dark image
- Active vs inactive user

Distance

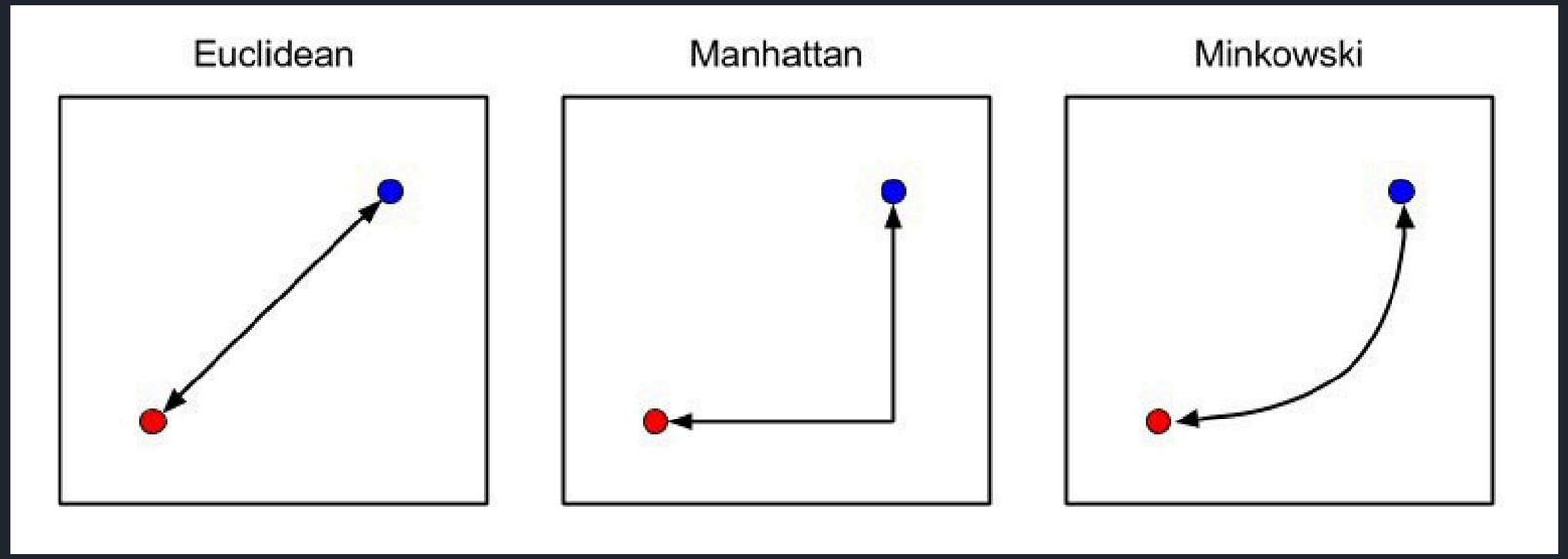
Distance between two vectors = similarity



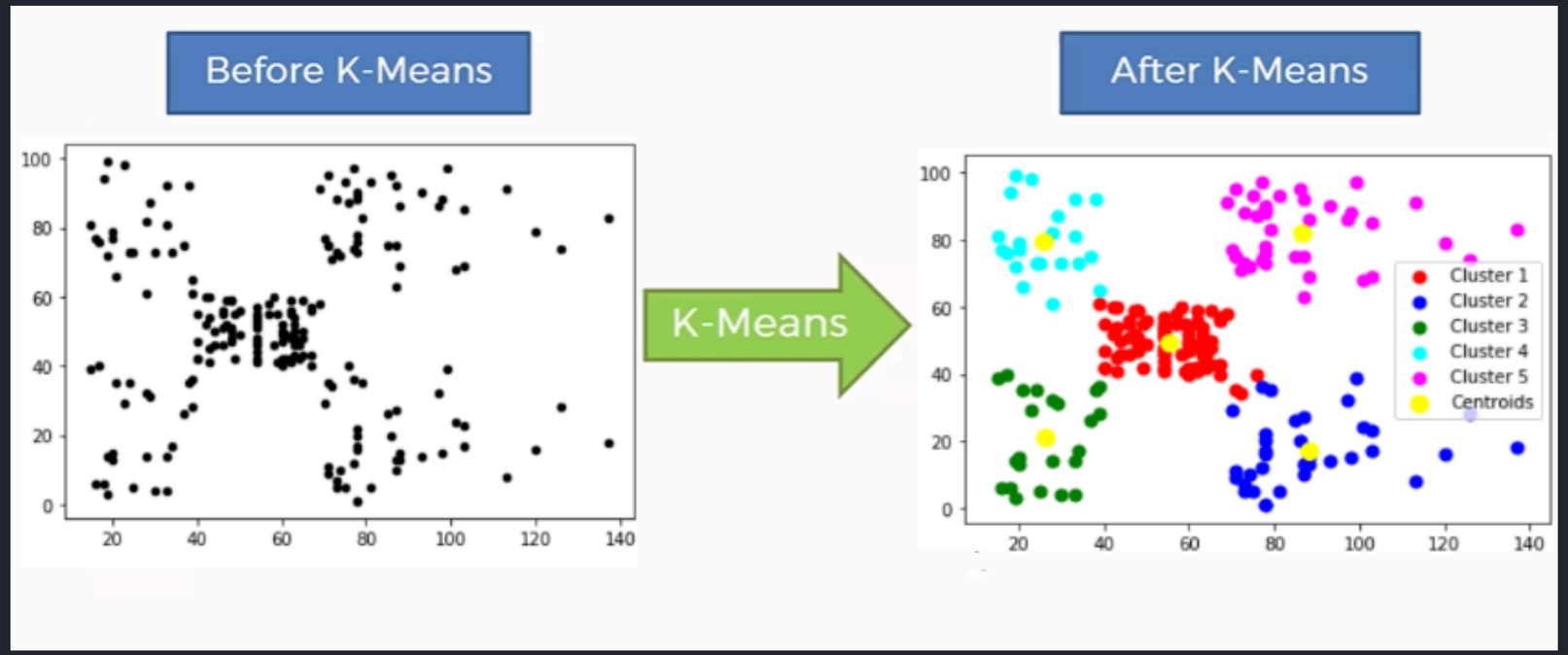
Close → similar
Far → different

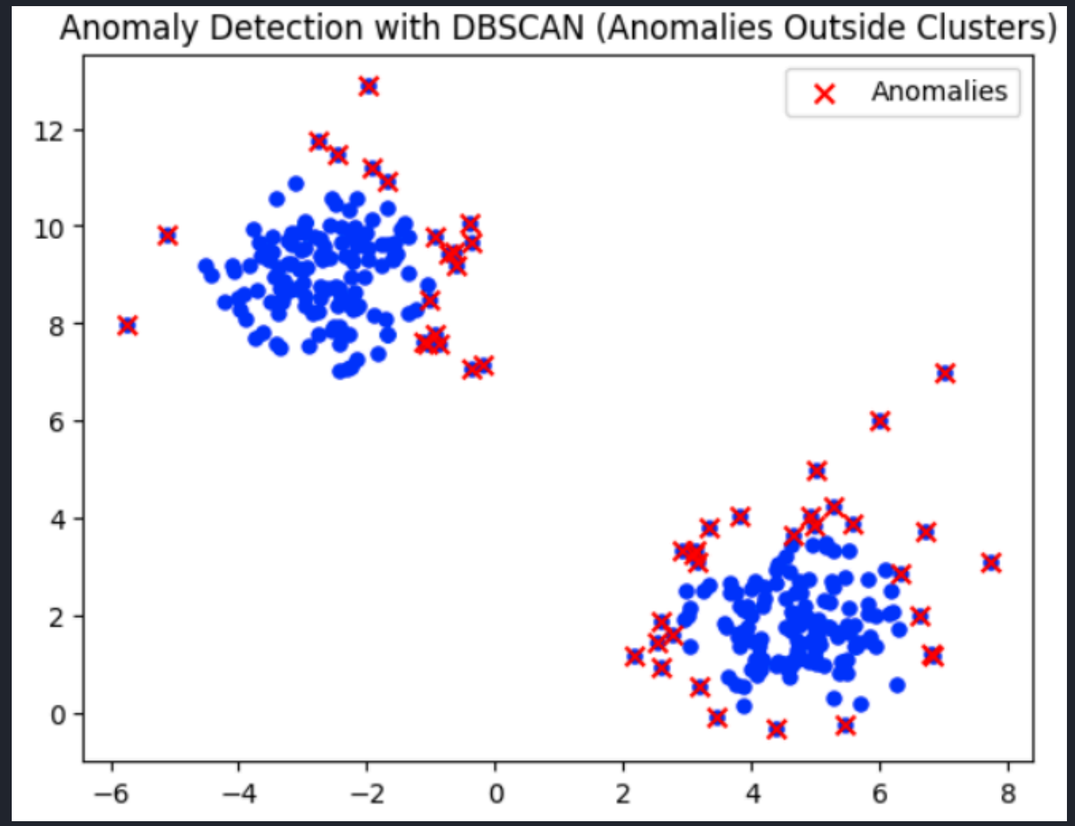
- KNN → compare distances
- Clustering → group close points
- Anomalies → very far points

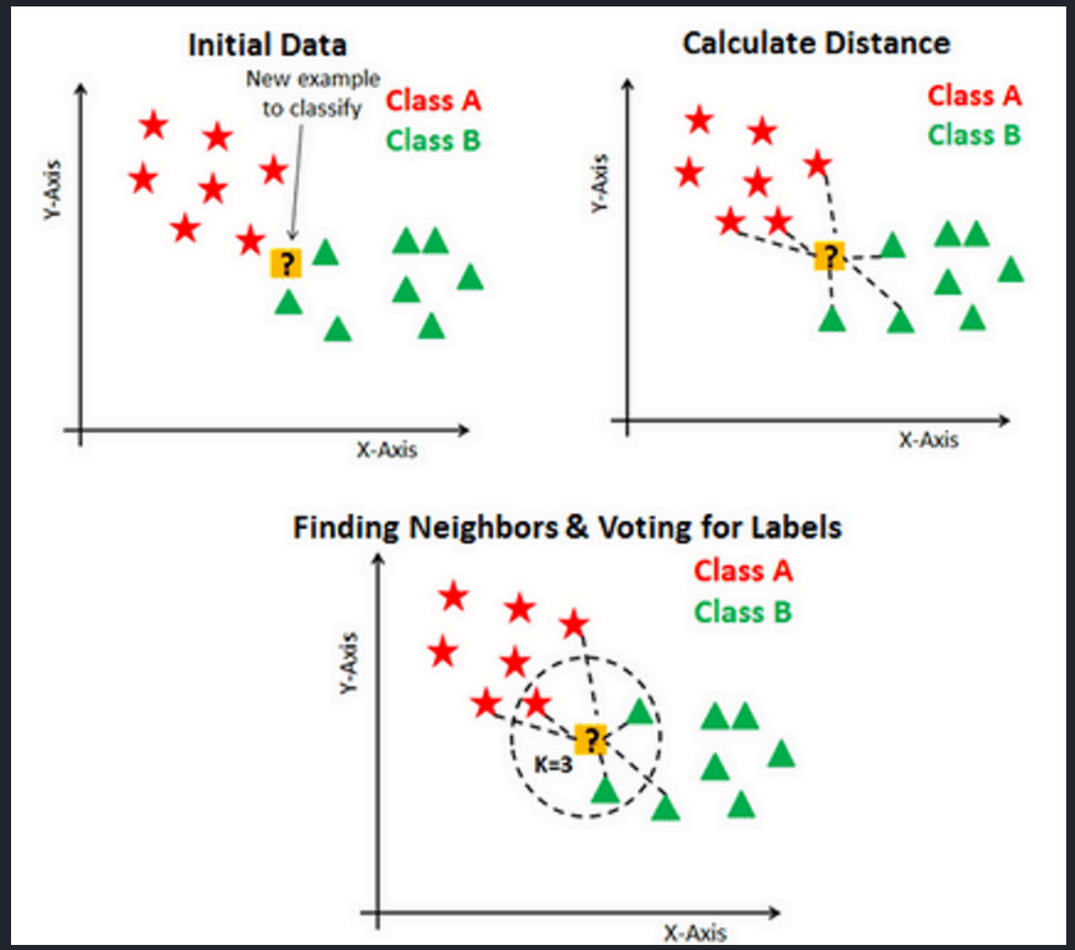
Types of Distance



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Before K-Means
After K-Means
K-Means
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Centroids


Anomaly Detection with DBSCAN (Anomalies Outside Clusters)
Anomalies
12
10
8
6
4
2
0
−6
−4
−2
0
2
4
6
8


Initial Data
New example
to classify
Class A
Class B
Y-Axis
X-Axis
?
Calculate Distance
Class A
Class B
Y-Axis
X-Axis
?
Finding Neighbors & Voting for Labels
Class A
Class B
Y-Axis
X-Axis
?
K=3

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03. Matrices

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What Is a Matrix?

```
matrix = collection of vectors
```



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	sepal length (cm)	sepal width (cm)	petal length (cm)	petal width (cm)	target
0	5.1	3.5	1.4	0.2	0
1	4.9	3.0	1.4	0.2	0
2	4.7	3.2	1.3	0.2	0
3	4.6	3.1	1.5	0.2	0
4	5.0	3.6	1.4	0.2	0
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	2
146	6.3	2.5	5.0	1.9	2
147	6.5	3.0	5.2	2.0	2
148	6.2	3.4	5.4	2.3	2
149	5.9	3.0	5.1	1.8	2

150 rows × 5 columns

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Rows and Columns

Rows → samples
Columns → features



	Titanic	Inception	Matrix	Avatar
User 1	5	4	1	3
User 2	2	5	4	1
User 3	4	3	5	2
User 4	1	2	2	5

Each row = one user

Each column = one movie

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Shape

```
Shape = (number of rows, number of columns)
```



Matrix Shape

$$X \in \mathbb{R}^{m \times n}$$

Example:

$$X \in \mathbb{R}^{100 \times 10}$$

Interpretation:

- Rows = 100 samples
- Columns = 10 features

Matrices in AI

📌 In almost all classical ML and deep learning models, the model is literally:

> a matrix of numbers learned from data

General form:

$$y = X * W$$

Where:

- X = data matrix (inputs)
- W = weight matrix (the model)
- y = output (predictions)

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What a weight matrix really means

📌 Each column of W answers:

How important is each feature for this output?

📌 Each row of W answers:

How does this input contribute to all outputs?

So learning = adjusting numbers inside W

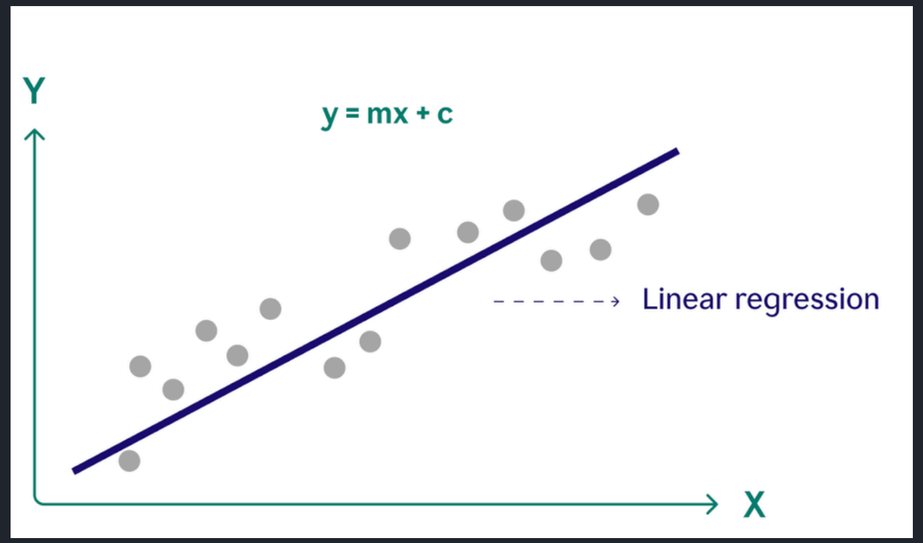
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1. Linear Regression

$$y = X * W$$

Where:

- W = weight vector / matrix
- Training = finding best W



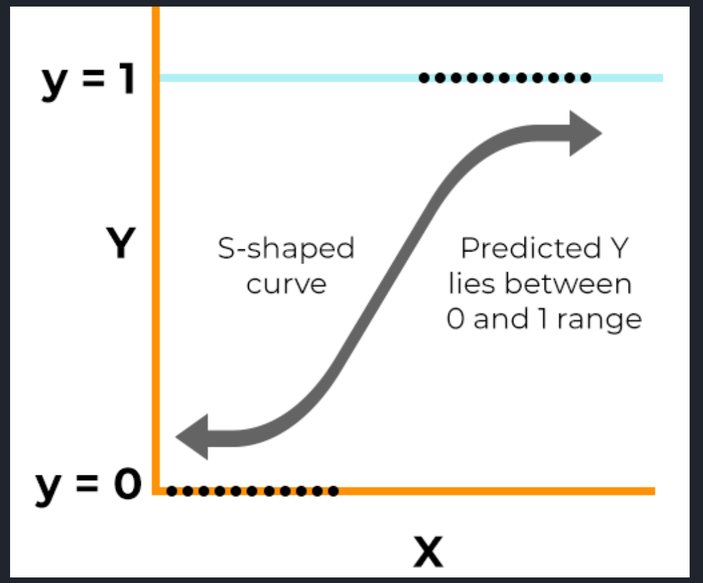
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2. Logistic Regression

$$y = sigmoid(X * W)$$

Where:

- core model = matrix W
- only difference = non-linear output



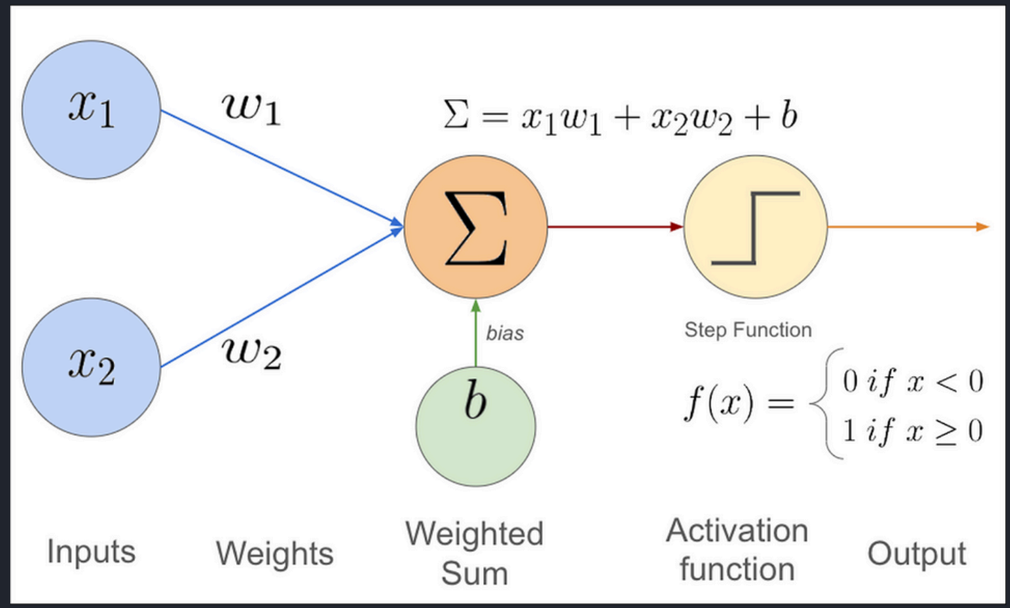
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3. Perceptron

$$y = Wx + b$$

Where:

- W = row vector
- This is a 1-layer neural network



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04. BASIC OPERATIONS

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Why Operations Matter

Operations answer:
What happens to data inside a model?

👉 We only care about meaning, not formulas

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Addition

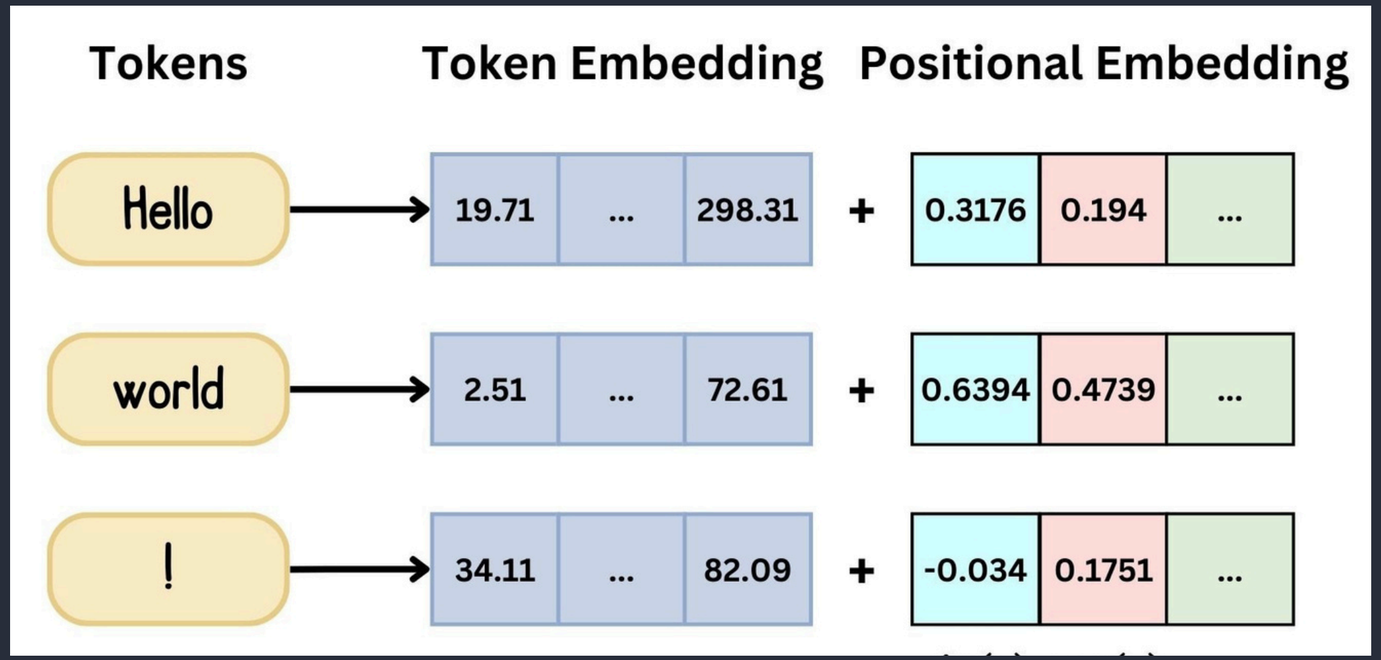
Addition = information fusion

In AI:

- merging signals
- combining embeddings
- adding bias to layers

Every time you see “+” in a model it means :

“inject more information”


Tokens
Token Embedding
Positional Embedding
Hello
19.71
...
298.31
+
0.3176
0.194
...
world
2.51
...
72.61
+
0.6394
0.4739
...
!
34.11
...
82.09
+
-0.034
0.1751
...

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Scalar Multiplication

Multiply by a number = amplify or reduce signal

Example:

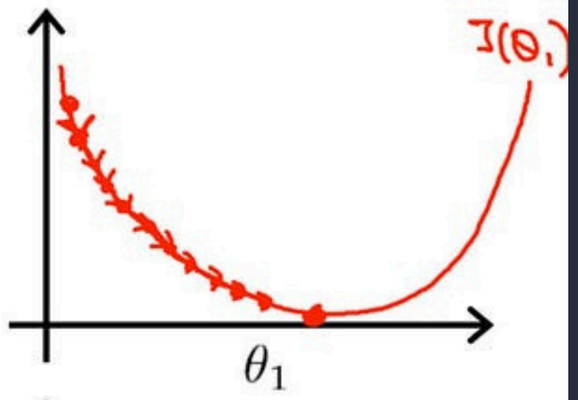
- Volume control
- Learning rate effect

Interpretation:
Same direction, different strength

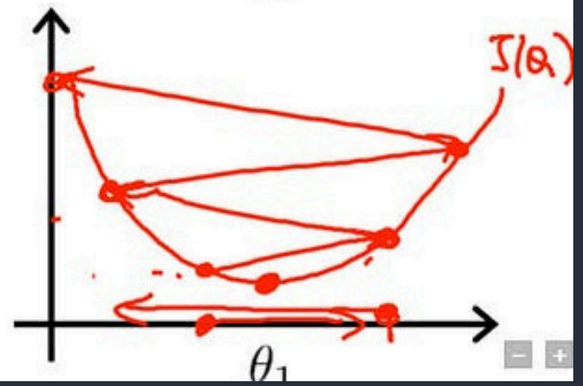
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$$\theta_1 := \theta_1 - \alpha \frac{\partial}{\partial \theta_1} J(\theta_1)$$

If α is too small, gradient descent can be slow.



If α is too large, gradient descent can overshoot the minimum. It may fail to converge, or even diverge.

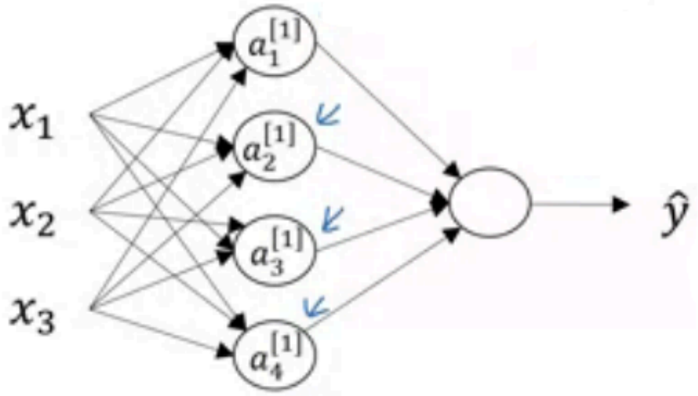


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Matrix × Vector

Input vector → Weight matrix → Output vector



$$z_1^{[1]} = w_1^{[1]T} x + b_1^{[1]};\ a_1^{[1]} = \sigma(z_1^{[1]})$$

$$z_2^{[1]} = w_2^{[1]T} x + b_2^{[1]};\ a_2^{[1]} = \sigma(z_2^{[1]})$$

$$z_3^{[1]} = w_3^{[1]T} x + b_3^{[1]};\ a_3^{[1]} = \sigma(z_3^{[1]})$$

$$z_4^{[1]} = w_4^{[1]T} x + b_4^{[1]};\ a_4^{[1]} = \sigma(z_4^{[1]})$$

$$z^{[1]} = \begin{bmatrix} — w_1^{[1]T} — \\ — w_2^{[1]T} — \\ — w_3^{[1]T} — \\ — w_4^{[1]T} — \end{bmatrix}_{(4,3)} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} b_1^{[1]} \\ b_2^{[1]} \\ b_3^{[1]} \\ b_4^{[1]} \end{bmatrix} = \begin{bmatrix} w_1^{[1]T} x + b_1^{[1]} \\ w_2^{[1]T} x + b_2^{[1]} \\ w_3^{[1]T} x + b_3^{[1]} \\ w_4^{[1]T} x + b_4^{[1]} \end{bmatrix} = \begin{bmatrix} z_1^{[1]} \\ z_2^{[1]} \\ z_3^{[1]} \\ z_4^{[1]} \end{bmatrix}$$

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Any Questions ?

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Thank you for your attention ✨

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